

URL Shortener API - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The project is a URL Shortener API designed to transform long URLs into shortened versions. While the current input is a quality validation checklist rather than a full functional specification, it identifies a target architecture based on FastAPI and SQLite. The core value proposition focuses on user-centric business needs and measurable success criteria, though the detailed requirements remain encapsulated in an external 'spec.md' file.

1.2 Maturity Assessment

The project state is ****BLOCKED****. Although the checklist indicates a 100% completion score for the validation process, the actual architectural graph is empty (1 node, 0 edges) because the provided document is a governance checklist and not the technical specification itself. Critical structural gaps exist in Goals, Functional Requirements, Non-Functional Requirements, and Scope, rendering the project unexecutable until the 'spec.md' source is ingested.

1.3 Technical Stack

- FastAPI
- SQLite

1.4 Architectural Constraints

- Technology-agnostic success criteria.
- Strict separation between business requirements and implementation details.

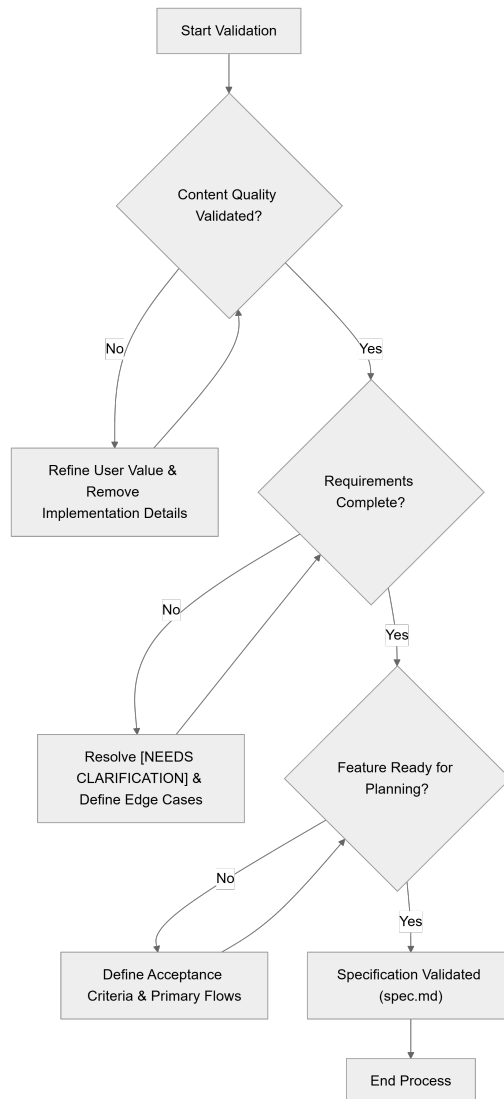
1.5 Critical Dependencies

- Availability of the external 'spec.md' document for functional requirement extraction.
- Integration between FastAPI runtime and SQLite database.

2. Architecture Workflows & Visual Diagrams

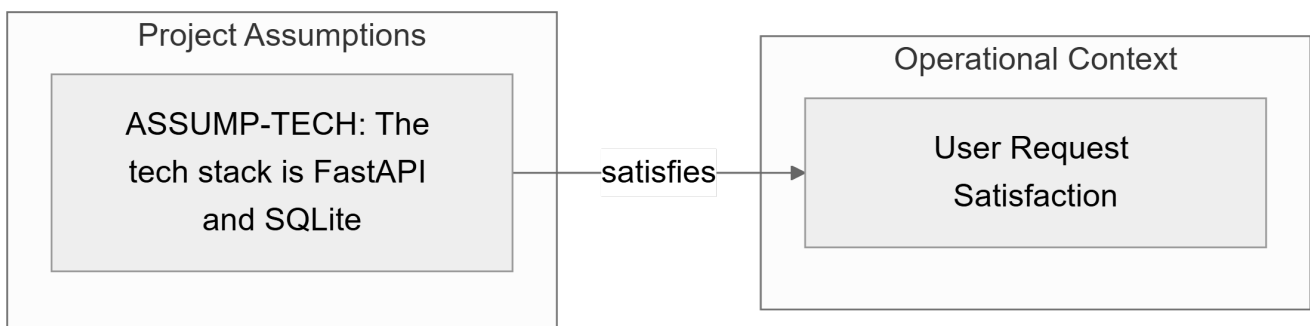
2.1 Specification Validation Workflow

Models the quality assurance process for the URL Shortener API specification based on the provided checklist criteria.



2.2 Project Assumptions Mapping

Traceability map for the technical assumptions identified in the quality checklist.



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section
ASSUMP-TECH	Assumption	The tech stack is FastAPI and SQLite	Notes

3.2 Security Rules

No security rules defined in the provided source data.

3.3 Data Models

No data models defined in the provided source data.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	Le document est une checklist; veuillez fournir le fichier 'spec.md' mentionné pour extraire les objectifs.
Functional Requirements	HIGH	Le document confirme que les exigences existent mais ne les liste pas. Fournir le document de spécifications source.
Non-Functional Requirements	HIGH	Le document confirme la complétude mais n'énonce pas les exigences non-fonctionnelles.
Scope & Out-of-Scope	HIGH	Fournir la définition du périmètre contenue dans le document de spécifications.
Open Questions & Uncertainties	MEDIUM	Bien que la checklist indique qu'il n'y a plus de marqueurs [NEEDS CLARIFICATION], les questions ouvertes initiales ne sont pas listées ici.

4.2 Remediation & Workflow

The project must transition from a "Validation" state to an "Implementation" state by ingesting the `spec.md` file. The current documentation serves as a quality gate; once the functional source is provided, the Requirements Traceability (Section 3.1) must be expanded to include all functional and non-functional requirements.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	Content Quality	The primary technical interface facilitating requests and responses for the link redirection system.
Feature	BUSINESS_DOMAIN	Feature Readiness	A distinct unit of functional capability that must satisfy specific measurable outcomes and acceptance criteria.